

Machine Learning Fundamentals

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data.

Types of Machine Learning

1. Supervised Learning

Uses labeled data.

Examples:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest

2. Unsupervised Learning

Uses unlabeled data.

Examples:

- K-Means Clustering
- PCA

3. Reinforcement Learning

Learns through rewards and penalties.

Model Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score

Overfitting occurs when a model performs well on training data but poorly on new data.

Cross-validation helps improve model performance.